

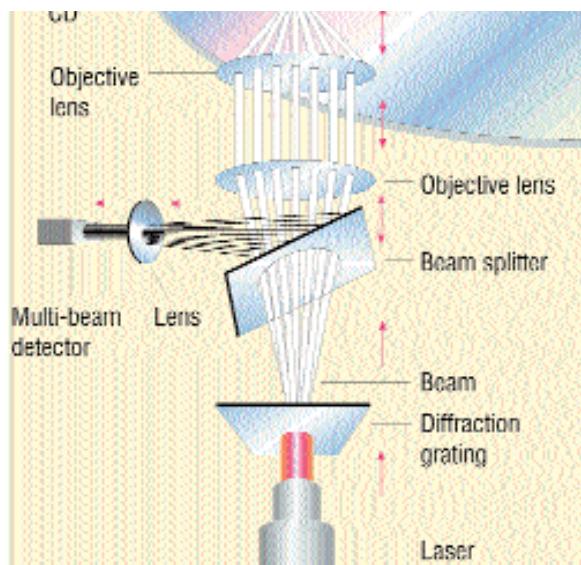
Spinning the Disc

Until just a few years ago Creative was the only brand in the CD-ROM drive market. The drives then cost around Rs 4,000, but prices have fallen steadily since then. A 56x rated CD-ROM drive now costs between Rs 1,700 and Rs 2,500. CD-ROM drives from ASUS and Acer are amongst the best performing with data transfer speeds ranging anywhere between 2.5 MBps to 4.7 MBps for the IDE models. SCSI CD-ROM drives, though available, are not something that most mainstream users need to consider.

In the case of DVD drives, there is a wide variation in prices and performance. Speeds range from 6x to 16x and the differences in transfer rates are very significant. The average cost of a DVD drive is about Rs 5,500 for a 12x model. DVD drives are usually adopted by users who have easy access to DVD movies.

TECHNOLOGY INSIGHT

A CD-ROM drive consists of a laser that is mounted on an assembly that lets it align itself onto any part of the surface of the disc as it spins. This laser reads the data which is stored in the form of a pit (0) and land (1). Additionally, there is a circuitry that performs the task of on-the-fly error correction which helps prevent errors



frequently encountered when reading scratched CDs. DVD drives differ basically due to the fact that the information density of the media is much higher.

SMART TIPS

Speed: In CD-ROM drives 1x corresponds to a data transfer rate of 150 KBps. Therefore, a 52x CD-ROM drive has a theoretical peak data transfer rate of $56 \times 150 \text{ KBps} = 8.2 \text{ MBps}$

Build quality: Some of the cheaper CD-ROM drives have very fragile loading and unloading trays. This tends to fail over a period of time. Look for a drive with sturdy mechanisms.

Also Remember...

- A notch or clasp in the drive tray that holds the media in place is a good idea if you are going to mount the drive vertically.
- Compatibility issues can come up with some CD-ROM drives which will create problems when reading from certain types of media such as CD-RW disks or other media that does not conform to the Red Book standard.

	Basic functionality	Entertainment enthusiast
You need	To use the drive to install software, play music and occasionally watch VCDs	To watch high quality movies as well as have the regular CD-ROM functionality
Look for	The fastest CD-ROM drive that you can afford	Look for a DVD-ROM that is rated at 12x or better
Our pick	ASUS CDS500	ASUS E612
Price range	Up to Rs 2,500	Rs 5,000 and above

